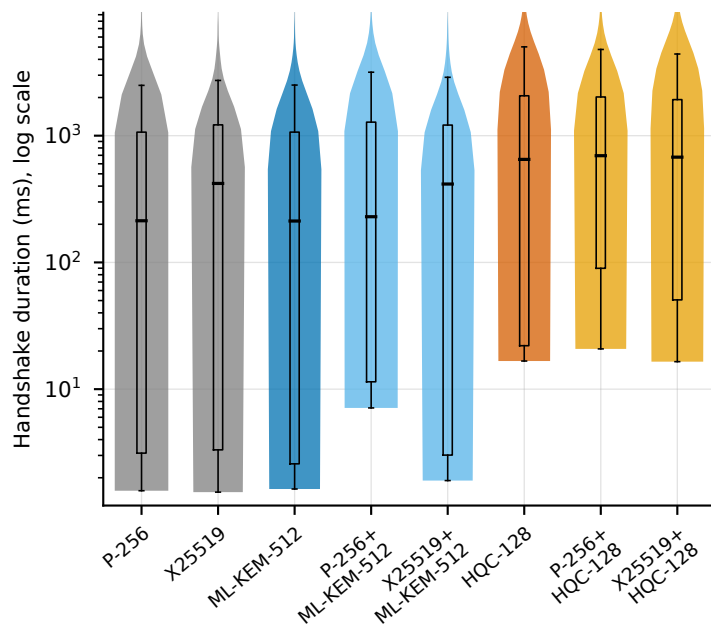
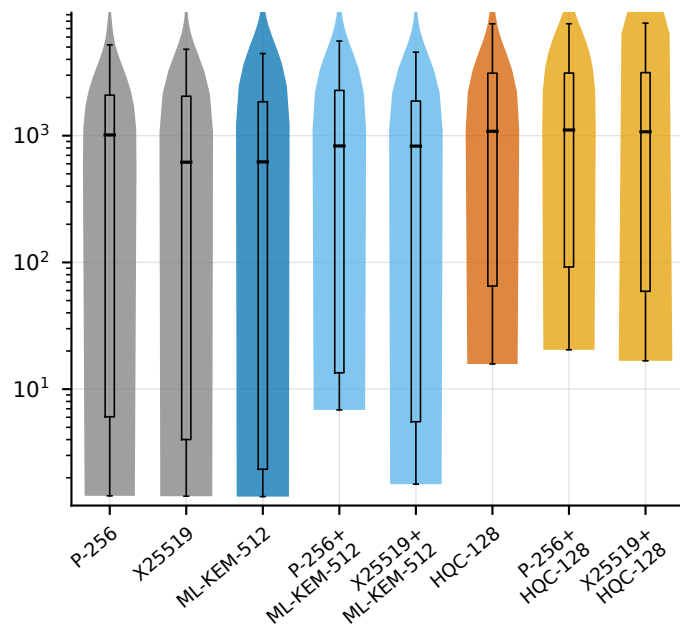


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	ed25519 Fail	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)	mldsa44 Fail
P-256	1.43k	412.0	104.43k	0	2.47k	250.6	56.93k	2/500
X25519	1.25k	257.7	55.74k	0	2.80k	299.3	105.40k	0
ML-KEM-512	1.18k	321.2	53.61k	1/500	3.17k	341.8	122.04k	2/500
P-256+ML-KEM-512	1.63k	403.7	106.52k	3/500	3.01k	309.1	108.50k	3/500
X25519+ML-KEM-512	1.15k	277.7	52.51k	0	2.51k	333.7	108.43k	2/500
HQC-128	2.82k	345.0	108.79k	2/500	3.62k	237.9	108.55k	1/500
P-256+HQC-128	2.28k	310.0	106.09k	1/500	4.50k	305.4	112.24k	0
X25519+HQC-128	2.94k	372.4	111.00k	1/500	5.24k	579.3	628.38k	1/500